

Eczema Prevention
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Aging fundamentally alters the skin's barrier function and immune response, making elderly individuals particularly vulnerable to eczematous conditions. Key changes include:

- Reduced ceramide and natural moisturizing factor (NMF) production, leading to transepidermal water loss (TEWL)
- Decreased sebaceous and sweat gland activity, reducing natural emolliency
- Thinning of the stratum corneum and dermis, impairing repair mechanisms
- Dysregulation of the Th2 immune axis, lowering the threshold for allergic sensitization
- Impaired Langerhans cell density, altering antigen-presenting capacity

Subtype	Key Features	Common Triggers in Elderly
Asteatotic eczema (eczema craquelé)	Dry, cracked, "crazy paving" pattern	Low humidity, harsh soaps, diuretics
Nummular dermatitis	Coin-shaped plaques, intensely pruritic	Dry environments, infections
Contact dermatitis (irritant)	Localized redness, scaling at exposure site	Soaps, incontinence products, topical medications
Contact dermatitis (allergic)	Vesicles, weeping at contact site	Fragrance, metals (nickel), rubber compounds
Seborrheic dermatitis	Greasy scales on scalp, face, chest	Malassezia overgrowth, neurological conditions
Stasis/varicose eczema	Lower leg involvement, pigmentation	Chronic venous insufficiency

Elderly patients may experience atypical presentations — reduced itch sensation, erythema obscured by chronic sun damage, or eczema mimicking other dermatoses (psoriasis, drug eruptions, cutaneous lymphoma). Accurate diagnosis before preventive regimen adoption is therefore essential.

Daily Skin Care & Hygiene Habits

Proper bathing technique is the cornerstone of eczema prevention in elderly individuals:

- Frequency: Bathe or shower once daily; more frequent bathing strips natural skin oils
- Water temperature: Use lukewarm water (27–32°C / 80–90°F); hot water aggravates barrier disruption and pruritus
- Duration: Limit bathing to 5–10 minutes to minimize TEWL
- Technique: Pat skin gently with a soft cotton towel — never rub
- Soak-and-seal method: Apply moisturizer within 3 minutes of exiting the bath to lock in moisture

Cleansing Product Selection

Recommended	Avoid
Fragrance-free, pH-balanced cleansers (pH 4.5–5.5)	Deodorant or antibacterial soaps
Syndets (synthetic detergent bars)	Soaps with sodium lauryl sulfate (SLS)
Gentle micellar water for sensitive areas	Products containing fragrance, alcohol, or colorants
Colloidal oatmeal-based cleansers	Loofahs and abrasive sponges

Skin folds (groin, axillae, beneath pendulous breasts, perianal area) require special attention in elderly patients with limited mobility:

- Ensure these areas are thoroughly dried after washing

- Apply a fragrance-free barrier cream or zinc oxide paste if friction or incontinence is present
- Avoid talcum powder; consider cornstarch-free medical powders only if recommended by a dermatologist

Moisturization Strategies

Choose emollients based on severity and tolerance:

1. Ointments (e.g., petrolatum/white soft paraffin): Most occlusive; best for very dry or cracked skin; may feel greasy
2. Creams: Good balance of occlusion and cosmetic acceptability; suitable for daytime use
3. Lotions: Lighter; less effective for severe dryness

- Apply emollients at least twice daily — morning and after bathing
- Use generous quantities: suggest 250–500 g/week for adults with active eczema
- Apply using downward, hair-growth-direction strokes to prevent folliculitis
- For hands, reapply after every handwashing
- Use pump dispensers rather than pots to minimize microbial contamination

Key Ingredients to Seek:

- Ceramides (1, 3, 6-II): Restore barrier lipid layers
- Hyaluronic acid: Humectant for water retention
- Glycerin and urea (2–10%): Effective humectants; higher-concentration urea (up to 40%) may be used for hyperkeratotic areas
- Niacinamide (vitamin B3): Anti-inflammatory, supports barrier function
- Colloidal oatmeal (1%): FDA-recognized skin protectant with anti-inflammatory properties

Dietary Choices and Nutritional Support

The Mediterranean dietary pattern has the strongest evidence base for reducing systemic inflammation relevant to eczema:

- High consumption of vegetables, fruits, legumes, whole grains, and olive oil
- Moderate consumption of fish and poultry
- Low consumption of red meat, processed foods, refined sugars, and trans fats

Key Nutrients and Their Roles:

Omega-3 Fatty Acids

- Inhibit pro-inflammatory eicosanoid synthesis
- Sources: Fatty fish (salmon, mackerel, sardines), walnuts, flaxseeds, chia seeds
- Target: 2–3 portions of oily fish per week, or supplementation after medical consultation

Vitamin D

- Supports innate antimicrobial peptide production (cathelicidins and defensins) critical for skin barrier
- Many elderly individuals are deficient due to reduced sun exposure and impaired cutaneous synthesis
- Dietary sources: Fortified dairy, egg yolks, oily fish
- Supplementation: Consider 800–2000 IU/day; assess serum 25(OH)D levels with a physician

Vitamin E

- Potent lipid-soluble antioxidant that reduces oxidative stress in skin cells

- Sources: Almonds, sunflower seeds, avocado, spinach

Zinc

- Essential cofactor for skin repair enzymes; deficiency linked to dermatitis
- Sources: Lean beef, legumes, pumpkin seeds, fortified cereals

Probiotics and Prebiotics

- Emerging evidence suggests gut microbiome modulation can shift immune responses away from Th2 polarization
- Probiotic-rich foods: Yogurt (with live cultures), kefir, fermented vegetables (kimchi, sauerkraut)
- Prebiotic foods: Garlic, onions, leeks, bananas, oats, asparagus

Nutritional Considerations Specific to Elderly Adults:

- Protein adequacy: Wound healing and skin repair require adequate protein intake (≥ 1.0 – 1.2 g/kg body weight/day)
- Caloric sufficiency: Undernutrition accelerates barrier breakdown; unintentional weight loss warrants clinical review
- B vitamins (B2, B3, B6, B12): Important for skin cell turnover; deficiencies are common in elderly, especially those on metformin or proton pump inhibitors

Hydration

- Minimum fluid intake: 1.5–2 litres of water daily (unless restricted by cardiac or renal conditions)
- Elderly individuals have reduced thirst perception — scheduled fluid intake (e.g., a glass of water at each meal and between meals) is more effective than relying on thirst
- Humidifiers: Indoor air humidity of 40–60% reduces skin dehydration; use HEPA-filtered humidifiers and clean weekly to prevent mold
- Limit caffeinated and diuretic beverages (coffee, tea, alcohol) that contribute to systemic dehydration

Environmental and Home Controls

Allergen	Control Strategies
House dust mite	Allergen-impermeable mattress/pillow covers; wash bedding weekly at $\geq 60^{\circ}\text{C}$; minimize carpets and heavy drapes
Pet dander	Limit pet access to bedroom; use HEPA air purifiers
Mold	Control indoor humidity (40–60%); use exhaust fans in bathrooms; inspect and remediate water-damaged areas
Pollen	Keep windows closed during high pollen counts; shower after outdoor exposure
Cockroach allergen	Seal food containers; address infestations promptly

- Use fragrance-free, dye-free laundry detergents and avoid fabric softeners
- Rinse laundry with an extra cycle to remove detergent residue
- Wear gloves for housework (cotton liners underneath rubber gloves to absorb sweat)
- Switch to non-irritant cleaning products (enzyme-based formulas)
- Install water softeners in hard-water areas — hard water has been associated with higher eczema prevalence

Temperature and Seasonal Adaptations

- Winter: Central heating drastically reduces indoor humidity; pair with a humidifier. Layer clothing to avoid sweating followed by rapid cooling
- Summer: Air conditioning also dries air; use humidifiers or indoor plants. Apply sunscreen (fragrance-free, mineral-based with zinc oxide or titanium dioxide) before sun exposure, and reapply every 2 hours
- Transitions: Skin needs 1–2 weeks to adapt to seasonal changes; increase moisturizer frequency during transitions

Clothing and Textile Selection

1. First choice: 100% cotton, bamboo-derived fabrics, Tencel/lyocell — breathable, moisture-absorbing, minimally irritating
 2. Second choice: Smooth, fine-gauge merino wool (not traditional wool); silk (for sensitive individuals who tolerate it)
 3. Avoid: Synthetic fabrics (polyester, nylon) that trap heat and moisture; coarse or rough-textured wool; tight elastic bands directly on skin
- Wash new clothes before first wearing to remove formaldehyde and finishing chemicals
 - Choose loose-fitting garments to minimize friction and allow air circulation
 - Wear seamless or inside-out clothing if seams cause irritation
 - Remove labels if they cause scratching
 - Layering is preferable to single heavy garments to manage body temperature and reduce sweating
 - Use fragrance-free, dye-free laundry products